

CIP-B102 Foundations of Computer and Digital Forensics

Lab 7: Steganography and Hidden Data Detection

Practical Lab | Steghide | LSB Concepts | Extraction | Password Dictionary Testing | Reporting

Important pre-lab instruction

Trainees must go through the Steganography slide walkthrough completely before starting this practical lab. The slide explains the concept of hiding information inside files, insertion/substitution, least significant bits (LSB), bitmap images, steghide usage, extraction, password dictionary preparation, and stegseek. Do not begin the lab until the materials have been read and understood.

1. Lab Summary

Course	CIP-B102 Foundations of Computer and Digital Forensics
Lab Title	Lab 7: Steganography and Hidden Data Detection
Estimated Duration	2-3 hours
Difficulty Level	Beginner to Intermediate
Required Platform	Kali Linux, Ubuntu, or any Linux VM with internet access
Core Tools	steghide, stegseek, wordlists/rockyou.txt, xxd, strings, file, sha256sum/md5sum
Submission Format	One PDF report plus the generated evidence files requested in this lab

2. Scenario

You are a junior digital forensic trainee. During a basic evidence review, an image file is suspected to contain hidden information. Your task is to understand how steganography works, hide a small evidence note inside a carrier image, extract it using the correct password, and then test a simple password dictionary attack in a controlled lab environment.

Authorization and safety rule

This lab is for ICDFA training only. Use only the lab image and files that you created yourself. Do not hide illegal content, do not test against third-party files without permission, and do not use these techniques against real evidence unless you are authorized and supervised.

3. Learning Objectives

- Explain what steganography means and how it differs from encryption.
- Describe insertion and substitution-based hiding methods, including the role of least significant bits (LSB).
- Prepare a carrier image and a secret text file for a controlled steganography experiment.
- Use steghide to embed and extract a file from a carrier image.

- Generate and compare hashes before and after embedding/extraction.
- Use a small wordlist and stegseek to test whether a weak steghide password can be recovered.
- Document all actions in a digital forensics report with screenshots and evidence notes.

4. Required Lab URLs and Resources Extracted from the Slide

Resource	Purpose	Exact URL / Command	Notes
Lab BMP image file	Carrier image for the steganography lab	https://drive.google.com/file/d/1CE-4EOe9-HEF_9mg3Y4N7u15qUblDm0/view?usp=sharing	Open in a browser and download. If command-line download is needed, use the direct download format shown below.
Direct download format for Google Drive file	Optional command-line download	https://drive.google.com/uc?export=download&id=1CE-4EOe9-HEF_9mg3Y4N7u15qUblDm0	If Google Drive asks for confirmation, download manually using the browser.
Stegseek project	Fast steghide password recovery tool	https://github.com/RickdeJager/stegseek	Use this for installation guidance or tool reference.
Kali repository line shown in slide	Required if Kali sources need non-free/contrib packages	<code>deb http://http.kali.org/kali kali-rolling main non-free contrib</code>	Only edit sources.list if packages are not available and you understand the impact.

5. What Students Need Before Starting

- A working Linux machine or VM. Kali Linux is recommended because it already supports many forensic tools.
- Internet access to download the carrier image and install required packages.
- A terminal with sudo access.
- At least 1 GB free disk space for the lab working folder.
- The Steganography slide material already studied before starting the practical tasks.

6. Tool Installation and Verification

Run the following commands in Kali/Ubuntu. Capture screenshots showing successful installation and tool versions where applicable.

```
sudo apt update
sudo apt install -y steghide file binutils xxd coreutils wget nano
sudo apt install -y wordlists

# If rockyou.txt is compressed, decompress a copy for the lab.
sudo gzip -dk /usr/share/wordlists/rockyou.txt.gz 2>/dev/null || true
```

```
# Try installing stegseek from your package repository.
sudo apt install -y stegseek || echo "If stegseek is unavailable, use the GitHub URL in the resource
table."

# Verify core tools.
steghide --version
file --version
xxd -h | head
sha256sum --version | head -n1
```

Package troubleshooting

If stegseek is not available through apt, students should visit the GitHub project URL listed in this lab and follow the installation method suitable for their system. Students must document whichever installation method they used.

7. Evidence Handling Rules for This Lab

- Create a dedicated working folder for Lab 7 only.
- Do not overwrite your original carrier image. Always preserve the original downloaded image.
- Record hashes of the original carrier image, secret text file, stego image, and extracted file.
- All commands must be captured in screenshots or copied into the command log section of your report.
- The final report must explain what each command did and why it was used.

8. Part A - Prepare the Working Folder and Download the Carrier Image

Create a lab folder and download the BMP image supplied in the slide. If the direct download command does not work because of Google Drive restrictions, open the original Google Drive URL in a browser and save the file into the same folder.

```
mkdir -p ~/CIP-B102-Lab7-Steganography
cd ~/CIP-B102-Lab7-Steganography

# Option 1: Try direct command-line download.
wget -O tower_original_image_for_lab.bmp "https://drive.google.com/uc?export=download&id=1CE-4E0e9-
HEF_9mg3Y4N7u15qUbIldm0"

# Option 2: If wget downloads an HTML page instead of the image,
# open this URL in a browser and download the file manually:
# https://drive.google.com/file/d/1CE-4E0e9-HEF_9mg3Y4N7u15qUbIldm0/view?usp=sharing

ls -lh
file tower_original_image_for_lab.bmp
sha256sum tower_original_image_for_lab.bmp > lab7_hashes.txt
cat lab7_hashes.txt
```

Required screenshot evidence for Part A:

- Working directory path.
- Downloaded image visible in the directory listing.
- `file` command output confirming the file type.

- SHA-256 hash of the original image.

9. Part B - Create the Secret Evidence Text File

Create a small text file that will be hidden inside the carrier image. Use your own name in the file content.

```
cat > secret.txt << 'EOF'
ICDFA CIP-B102 Lab 7 Evidence Note
Student Name: FIRSTNAME LASTNAME
Message: This file was hidden and recovered during a controlled steganography lab.
EOF

cat secret.txt
file secret.txt
sha256sum secret.txt >> lab7_hashes.txt
cat lab7_hashes.txt
```

Required screenshot evidence for Part B:

- Content of `secret.txt` displayed in the terminal.
- Hash of `secret.txt` appended to `lab7\_hashes.txt`.

10. Part C - Hide the Secret File Using Steghide

Use steghide to embed the secret file into the BMP carrier image. Use the lab password `1234` so that the password recovery section can be completed later.

```
# Embed the secret file into a new stego image.
steghide embed \
  -ef secret.txt \
  -cf tower_original_image_for_lab.bmp \
  -sf tower_stego_lab7.bmp \
  -p 1234

# Verify the new stego file.
ls -lh tower_original_image_for_lab.bmp tower_stego_lab7.bmp
file tower_stego_lab7.bmp
sha256sum tower_stego_lab7.bmp >> lab7_hashes.txt
cat lab7_hashes.txt
```

Required screenshot evidence for Part C:

- The steghide embed command.
- Directory listing showing original carrier image and stego image.
- Hash of the stego image.
- A short explanation of why the original image hash and stego image hash are different.

11. Part D - Extract the Hidden File Using the Correct Password

Extract the hidden evidence file from the stego image and verify that the recovered file is identical to the original secret file.

```
mkdir -p extracted
steghide extract \
-sf tower_stego_lab7.bmp \
-xf extracted/extracted_secret.txt \
-p 1234

cat extracted/extracted_secret.txt
sha256sum extracted/extracted_secret.txt >> lab7_hashes.txt

# Verify that the original and extracted files are the same.
diff secret.txt extracted/extracted_secret.txt && echo "MATCH: extracted file is identical"
sha256sum secret.txt extracted/extracted_secret.txt
```

Required screenshot evidence for Part D:

- Successful steghide extraction command.
- Recovered text content displayed in the terminal.
- `diff` or hash comparison proving the recovered file matches the original.

12. Part E - Basic Visual and Binary Comparison

Compare the original and stego images at a basic level. This section helps students understand that a visually similar image can still have different underlying data.

```
# Compare sizes.
ls -lh tower_original_image_for_lab.bmp tower_stego_lab7.bmp

# Compare hashes.
sha256sum tower_original_image_for_lab.bmp tower_stego_lab7.bmp

# Show first 64 bytes of both files.
xxd -l 64 tower_original_image_for_lab.bmp
xxd -l 64 tower_stego_lab7.bmp

# Search readable strings in the stego image.
strings tower_stego_lab7.bmp | head -n 20
```

Required screenshot evidence for Part E:

- File size comparison.
- Hash comparison.
- First 64 bytes of original and stego image using `xxd`.
- Short interpretation of whether visible appearance alone is enough to detect steganography.

13. Part F - Password Dictionary Test with Stegseek

In this controlled lab, students will test why weak steganography passwords are risky. Instead of using a huge dictionary first, create a small lab dictionary that contains the known password `1234`.

```
cat > lab7_wordlist.txt << 'EOF'
password
admin
icdfa
```

```

cybersecurity
1234
letmein
EOF

cat lab7_wordlist.txt

# Run stegseek against the lab stego image using the small wordlist.
# The exact output may vary depending on your installed stegseek version.
stegseek tower_stego_lab7.bmp lab7_wordlist.txt

# If stegseek extracts a file automatically, list the directory and document the result.
ls -lh

```

If stegseek fails on your system

Some environments may not have stegseek installed or may behave differently depending on the file format/tool version. If it fails, document the error message, confirm that steghide extraction works with the correct password, and include the GitHub URL from the resource table as the tool reference.

Required screenshot evidence for Part F:

- The custom `lab7\_wordlist.txt` content.
- The stegseek command and result.
- A short explanation of why weak passwords are dangerous in steganography.

14. Part G - Independent Student Task

Each student must create their own steganography evidence file and submit it with their report.

1. Create a new text file named `firstname\_lastname\_hidden\_note.txt`.
2. Write a short message explaining what steganography is and why it matters in digital forensics.
3. Embed the file into the lab BMP image using a password of your choice.
4. Record the password in your private notes, but do not put it on the first page of the report.
5. Extract the file again and prove it matches the original using hash comparison.
6. Create a mini dictionary that includes your chosen password and test stegseek again.
7. Include screenshots and explanation for every step.

15. Troubleshooting Guide

Problem	Likely Cause	Solution
Downloaded file is HTML, not BMP	Google Drive direct download confirmation or permission issue	Open the original Google Drive URL in a browser and download manually.
steghide: could not open cover file	Wrong filename or not in the correct folder	Run `pwd`, `ls -lh`, and confirm the exact file name.
steghide extraction fails	Wrong password or wrong stego file	Use the lab password `1234` for the guided task and confirm the stego file name.
rockyou.txt not found	wordlists package not installed or still	Install wordlists and decompress

	compressed	` /usr/share/wordlists/rockyou.txt.gz `.
stegseek not found	Package not installed or not available in repository	Try apt first; otherwise use the GitHub project URL listed in the resource section.

16. Final Report Structure

- Cover page: student name, course code, lab title, date, and operating system used.
- Pre-lab confirmation: a short statement that the slide walkthrough was studied before the lab.
- Tools used: steghide, stegseek, xxd, strings, hashes, and any other tools used.
- Download evidence: source URL used and screenshot of the downloaded file.
- Procedure: step-by-step commands with screenshots.
- Hash table: original image, secret file, stego image, extracted file.
- Findings: explain whether the hidden file was recovered successfully.
- Password test: explain the result of stegseek or document why it failed.
- Reflection: what steganography means in digital forensic investigations.
- Appendix: command log and any error messages.

17. Submission Checklist

Item	Required?	Student Confirmation
PDF lab report with screenshots	Yes	☐ Completed
Original secret.txt file	Yes	☐ Completed
tower_stego_lab7.bmp or student-created stego image	Yes	☐ Completed
Extracted hidden file	Yes	☐ Completed
Hash file: lab7_hashes.txt	Yes	☐ Completed
Custom wordlist used for password test	Yes	☐ Completed
Short explanation of LSB/substitution concept	Yes	☐ Completed

18. Marking Rubric - 100 Marks

Criteria	Marks	Assessment Guide
Pre-lab readiness and environment setup	10	Confirmed slide review, installed required tools, created working folder correctly.
Carrier image download and preservation	10	Used the correct lab URL, preserved original image, recorded hash.
Steghide embedding procedure	15	Successfully embedded secret file and documented the process with screenshots.

Extraction and verification	20	Successfully extracted hidden file and proved integrity with hashes/diff.
Binary/hash comparison and interpretation	10	Compared original and stego files and explained the forensic relevance.
Password dictionary/stegseek task	15	Created wordlist, tested stegseek or documented failure properly, explained weak password risk.
Report quality and screenshot evidence	15	Clear, well-structured, repeatable report with commands and screenshots.
Professional ethics and evidence handling	5	Followed authorized-use rules and did not overwrite original evidence.

19. Reflection Questions

8. What is steganography and how is it different from encryption?
9. Why are BMP images commonly used when explaining LSB steganography?
10. Why does the stego image hash change after embedding a file?
11. Can a normal visual inspection prove whether an image contains hidden data? Explain.
12. Why is password strength important when using steghide?
13. How can steganography be relevant in digital forensic investigations?

20. Instructor Notes

- The guided password for the first stego file is `1234` for teaching and verification only.
- Students may use their own password for the independent task, but they must demonstrate extraction and verification.
- Assess the report based on repeatability: another examiner should be able to follow the report and reproduce the same outcome.
- Students should not be penalized if Google Drive requires manual download, provided they document the process properly.
- If stegseek installation fails, students should document the failed installation and still complete the steghide embedding/extraction workflow.

21. References and Slide-Based Links

- Steganography Lab Image File: https://drive.google.com/file/d/1CE-4EOe9-HEF_9mg3Y4N7u15qUbIIIDm0/view?usp=sharing
- Stegseek GitHub Project: <https://github.com/RickdeJager/stegseek>
- Kali Package Repository Line: `deb http://http.kali.org/kali kali-rolling main non-free contrib`